

Pipeline Frozen, Reproducible

What was done

1. **Consolidated the scientific pipeline into one reproducible script** (`src/pipeline_end_to_end.py`) — this closes a real gap: the Day 3 fixes (timezone, per-night NCTL, fixed baseline, min-max normalization) only ever lived inside the day3 notebook, never as a standalone reusable script anyone could just run.
2. **Actually tested it end-to-end** in a clean environment (not just written and assumed correct): ran from raw `era5_merged.nc` all the way to `CTBI.csv` + sensitivity test. Output matched previously verified values exactly (baseline 32.15, CTBI range 0.089–0.754, sensitivity correlation ~0.95) — confirms the script is genuinely reproducible, not just plausible-looking code.
3. **No new features added** — per the Day 4 brief, this is a freeze. The script only reorganizes and consolidates already-validated logic (`process_weather.py`, `cos_solar.py`, `mrt.py`, `utci.py`) rather than changing any calculation.
4. **README updated** with a "How to Reproduce" section explaining exactly how to run the pipeline and what each output file contains.
5. **Future XGBoost architecture** — checked the existing README; it's already well-documented (## 🤖 Future AI/ML Extension section) with the correct honesty framing (SHAP explanations only shown once a real model produces them). No changes needed there.

Files produced

- `src/pipeline_end_to_end.py`
- README addendum (paste into `README.md`, ideally right after the existing "🏗️ Methodology" section)

Handoff

- This script is now the single source of truth for regenerating `CTBI.csv` — if anyone finds a genuine bug in the science going forward, fix it here, not in a notebook, so it doesn't silently drift out of sync again.

